

EN	<= Language Selection - English or French <= Sélection de la Langue - Anglais ou Français	
TYPE	SECTION	DESCRIPTION
Section	PROJECT REQUIREMENTS	
Requirement	GitHub Branch Structure	Your repository must follow this branching model throughout development: • main → Production-ready branch (only branch graded) • dev → Integration branch; receives completed features • feature/* → One branch per feature → Must be created from dev → Must be merged back into dev when completed No direct commits to main
Requirement	GitHub Final Branch State	Before submitting the module: 1. All feature branches must be merged into dev 2. The dev branch must then be merged into main 3. The main branch must reflect the final, stable version of the project The commit history must clearly demonstrate this workflow: • feature → dev → main Only the main branch will be graded.
Requirement	React Bootstrap Installed	Install React Bootstrap and actively use its components within your project.
Requirement	Postman Collection	Create and use a Postman collection that covers all endpoints for the current module. All parameters must be set in advance, and the collection should function without requiring any modification to the queries. Export the collection as PostmanCollection.json and place it in the root directory of your project.
Requirement	README.md Document	Create a README.md file that clearly explains your project to someone with no prior knowledge of the setup. It must include: • Project Title and clear Description • Tech Stack • Project Structure • Installation / Setup Instructions • Environment Variables • API Documentation • Author * If a section does not apply to your project, clearly mention that instead of leaving it unexplained. File name & Location : ./README.md
Requirement	Submission Summary Document	Submit a summary document (.txt, .md, .docx) through the platform during the deliverable step, as an attachment or public Google Docs link (view access enabled). The document must include: • Student Name • Module Name • Project Repository Link • All Required Video Links • Any Required Credentials (e.g., .env values, login details if applicable) File name: submission-summary.md / .txt / .docx Important: This file must NOT be committed to GitHub.
Section	AI-NATIVE PROJECT SPECIFICATION & IMPLEMENTATION REQUIREMENTS	
Sub-Section	Global Specifications	
Requirement	AI Specification document	Create a project-wide AI specification document : • Project identity + scope (in/out) • Architecture / repo structure • Allowed tech & constraints • Coding standards / conventions • Global "Definition of Done" + cross-feature rules IMPORTANT : Must be read first before implementing any feature. File name & Location : ./ai/ai-spec.md
Sub-Section	Feature - Home page	
Requirement	AI Feature Specification - Home Page	Create a feature specification document : • Feature goal + scope (in/out) • Requirements breakdown + user flow • Interfaces involved (pages/components/endpoints) • Data + validations + expected behavior • Acceptance criteria (how to verify it works) IMPORTANT: Must be used with the Global AI Spec. File name & Location : ./ai/features/home-page.feature.md
Requirement	Home Page - Create	The home page is created and set as the default view after login. It displays a grid of cards.
Requirement	Home Page - Agent Management Card	The Home page must include an Agent Management card that navigates to the agent table page.
Requirement	Home Page - Transaction Card	The Home page must include a Transaction card that navigates to the transaction page.
Sub-Section	Feature - Notifications	
Requirement	AI Feature Specification - Notifications	Create a feature specification document : • Feature goal + scope (in/out) • Requirements breakdown + user flow • Interfaces involved (pages/components/endpoints) • Data + validations + expected behavior • Acceptance criteria (how to verify it works) IMPORTANT: Must be used with the Global AI Spec. File name & Location : ./ai/features/notifications.feature.md

Requirement	Notifications - on Login	Both success and error notifications appear during login.
Requirement	Notifications - on Create Agent	Both success and error notifications appear when creating an agent.
Requirement	Notifications - on Update Agent	Both success and error notifications appear when updating an agent.
Requirement	Notifications - on Delete Agent	Both success and error notifications appear when deleting an agent.
Requirement	Notifications - on Submit Transaction	Both success and error notifications appear when submitting a transaction.
Sub-Section	Feature - Modals	
Requirement	AI Feature Specification - Modals	Create a feature specification document : • Feature goal + scope (in/out) • Requirements breakdown + user flow • Interfaces involved (pages/components/endpoints) • Data + validations + expected behavior • Acceptance criteria (how to verify it works) IMPORTANT: Must be used with the Global AI Spec. File name & Location : ./ai/features/modals.feature.md
Requirement	Modals - Create Agent	Confirmation modal appears before creating an agent.
Requirement	Modals - Update Agent	Confirmation modal appears before updating an agent.
Requirement	Modals - Delete Agent	Confirmation modal appears before deleting an agent.
Requirement	Modals - Submit Transaction	Confirmation modal appears before submitting a transaction.
Sub-Section	Feature - Session process	
Requirement	AI Feature Specification - Session Process	Create a feature specification document : • Feature goal + scope (in/out) • Requirements breakdown + user flow • Interfaces involved (pages/components/endpoints) • Data + validations + expected behavior • Acceptance criteria (how to verify it works) IMPORTANT: Must be used with the Global AI Spec. File name & Location : ./ai/features/session-process.feature.md
Requirement	Session Process - Saved on Login	When a login succeeds, a new session must be created via POST /session and stored in the Session collection in MongoDB, which must contain all required fields.
Requirement	Session Process - Token Generated	Each session generates a unique, valid token.
Requirement	Session Process - TTL (24 hours)	The session collection uses TTL so sessions expire automatically after 24 hours.
Requirement	Session Process - Endpoint Response Format	The POST /session endpoint returns valid data in the correct format.
Requirement	Session Process - Token Stored in Cookie	After session creation, the token is stored in a cookie.
Requirement	Session Process - Token Check on Navigation	Every page checks for a valid session_token.
Sub-Section	Feature - Validation process	
Requirement	AI Feature Specification - Validation Process	Create a feature specification document : • Feature goal + scope (in/out) • Requirements breakdown + user flow • Interfaces involved (pages/components/endpoints) • Data + validations + expected behavior • Acceptance criteria (how to verify it works) IMPORTANT: Must be used with the Global AI Spec. File name & Location : ./ai/features/validation-process.feature.md
Requirement	Validation Process - Validation Logic	Session validation must check in MongoDB whether the session exists and whether the token is valid.
Requirement	Validation Process - GET /validate_token Parameters	Endpoint correctly receives the token as a query parameter.
Requirement	Validation Process - GET /validate_token - Return Format	Endpoint returns properly formatted data.
Requirement	Validation Process - Redirect If No Token	If no token or invalid token is found, redirect the user to the login page.
Requirement	Validation Process - Redirect If Token Valid	If a valid session exists, redirect the user to the admin home page.
Sub-Section	Feature - Header bar	
Requirement	AI Feature Specification - Header Bar	Create a feature specification document : • Feature goal + scope (in/out) • Requirements breakdown + user flow • Interfaces involved (pages/components/endpoints) • Data + validations + expected behavior • Acceptance criteria (how to verify it works) IMPORTANT: Must be used with the Global AI Spec. File name & Location : ./ai/features/header-bar.feature.md
Requirement	Header Bar - Create Agent Button Placement	Move the "Create Agent" link from the header bar into the Agent Management component.

Requirement	Header Bar - Display Username	The logged-in user's name must appear in the navbar.
Sub-Section	Feature - Transactions process	
Requirement	AI Feature Specification - Transaction Process	Create a feature specification document : • Feature goal + scope (in/out) • Requirements breakdown + user flow • Interfaces involved (pages/components/endpoints) • Data + validations + expected behavior • Acceptance criteria (how to verify it works) IMPORTANT: Must be used with the Global AI Spec. File name & Location : <code>./ai/features/transaction-process.feature.md</code>
Requirement	Transactions Process - GET /transaction-data Returns Proper Data	Endpoint returns an array of the last 10 transactions in the correct format.
Requirement	Transactions Process - Form Saved to DB	Form submission saves the transaction via POST /transaction.
Requirement	Transactions Process - Transaction Object Structure	Must follow {date, amount, agent_id}.
Requirement	Transactions Process - POST /transaction Parameters	Endpoint accepts amount and agent_id in the body.
Requirement	Transaction Process - List Displayed	The transaction page is created and shows the last 10 transactions.
Requirement	Transaction Process - Table Columns	Table must display: date, amount, agent full name.
Requirement	Transactions Process - Sorted	Transactions are sorted by most recent first.
Requirement	Transactions Process - Transaction Amount - Positive Only	Amount input accepts positive numbers only.
Requirement	Transactions Process - Agent Dropdown	A dropdown menu lists all available agents by name and ID.
Section	TECHNICAL INTERVIEW REQUIREMENTS	
Requirement	CONCEPTS.md - Concept & Explanation Video	Part 1 - Create CONCEPTS.md and list 3 challenging concepts used in your project. For each concept, include: • Concept name • Purpose in the project • Why it was challenging • Usage location (file + line if applicable) File name & Location : <code>./CONCEPTS.md</code> Part 2 - Record a video (~ 5 - 10 minutes) and explain the 3 concepts in your own words to demonstrate your understanding. Part 3 - Upload to YouTube (Unlisted) and include the link in your submission summary document.
Requirement	Problem Solving - Results & Explanation Video	Part 1 - Demonstrate your understanding of integration, validation logic, testing mindset by solving the following problems: • Function Composition → https://leetcode.com/problems/function-composition/ • Memoize → https://leetcode.com/problems/memoize/ • Debounce → https://leetcode.com/problems/debounce/ • Allow One Function Call → https://leetcode.com/problems/allow-one-function-call/ • Climbing Stairs → https://leetcode.com/problems/climbing-stairs/ Part 2 - For each completed challenge: • Take a clear screenshot of your solution. • Save the screenshot in your project folder using the following structure: • File Name & Location: <code>./LeetCode-Challenges/<challenge-name>.png</code> Part 3 - In addition, record a problem-solving explanation video (~5–10 minutes) where you explain each challenge in your own words, focusing on the reasoning behind the solution, not just the final code or answer. Part 4 - Upload to YouTube (Unlisted) and include the link in your submission summary document.
Requirement	Technical Demo & Code Overview Video	Record a video (~ 5 - 10 minutes) divided into two parts: Part 1 - Functional Demonstration (UI & Behavior) Demonstrate how each requirement works by interacting with: • User Interface (UI) • Postman (API calls, if applicable) • DBever / MongoDB (Database, if applicable) Part 2 - Code Explanation (Internal Implementation) Explain how each requirement is implemented in the code, including: • Project structure • How each requirement works internally (state management, navigation, APIs, core logic) Part 3 - Upload to YouTube (Unlisted) and include the link in your submission summary document.
Section	DEADLINE REQUIREMENTS	
Requirement	Deliverable Submitted on Time	Submit the complete project no later than 11:59 PM on Friday of the assigned deadline week. Approved extensions allow extra time, but points will still be deducted for late submissions.
Section	EXTRA MILES	Extra miles will only be considered if all main requirements are fully completed and reviewed by a coach.
Extra Mile	Report Page & Visualizations	Create a Report page that includes a bar graph showing total transactions per agent and a line graph displaying daily totals for the past 14 days
Extra Mile	GET /report-data - Supplies All Chart Data	Implement the GET /report-data endpoint to supply all required chart data and return it in the format: { status: "ok", data: { agent_bar_data: [...], transaction_line_data: [...] }, message: null }.

Section	SOFT/PRO SKILLS	Communication, progress updates, and project reviews are mandatory in this program. You are expected to : → Respond to your coaches within 24 hours when contacted. → Provide at least two (2) progress updates per week. → Schedule at least one (1) project review per week before Friday. Failure to meet these requirements may result in a failed module.
ProDev	Attendance / Punctuality	• Does the student attend scrums or scheduled meetings (when applicable)? • Are they on time? • Do they respond to messages or check-ins within 24 hours?
ProDev	Autonomy & Initiative	• Does the student complete tasks without requiring frequent coach intervention? • Are they able to research, troubleshoot, or find solutions independently? • Do they push code or submit deliverables regularly?
ProDev	Professionalism	• Does the student behave reliably and respectfully in all formats (Slack, GitHub, Meet, etc.)? • Do they submit deliverables on time and follow instructions without being reminded?
ProDev	Communication & Collaboration	• Is the student capable of expressing their progress, blockers, or needs clearly? • Do they participate in discussions, answer questions when tagged, or collaborate when in teams?
ProDev	Technical Proficiency	• Does the student demonstrate solid understanding of concepts through their work? • Are their solutions well-written, complete, and using appropriate tools or methods?
ProDev	Attention to Detail	• Are there minimal spelling, syntax, or formatting errors? • Does the student follow naming conventions, indentation, and file structure guidelines?
ProDev	Going the Extra Mile	• Does the student add extra features, improvements, or bonus tasks not explicitly required? • Do they show signs of pride in craftsmanship and curiosity?